The Journal of Open Source Software

Py3plex 2.0: Declarative Workflows for Multilayer Network Science

Evgenija Popchanovska[2*], **Boshko Koloski**[1*], **Yana Melnyk**[4], **Benjamin Renoust**[3*], **and Blaž Škrlj**[1*]

1 Jožef Stefan Institute, Ljubljana, Slovenia **2** Faculty of Computer Science and Engineering, Skopje, North Macedonia **3** Institute for Datability Science Osaka University, Japan **4** Epitech - L'école de l'excellence informatique * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 27 October 2026
Published: unpublished

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Summary

Multilayer networks represent systems with several kinds of relationships. Keeping those relationships in separate layers allows researchers to study each layer and the connections between them (Boccaletti et al., 2014; Kivelä et al., 2014). Py3plex is a Python library for analyzing and visualizing these networks.

The original Py3plex publications focused on multilayer network representation, analysis, and visualization (Škrlj et al., 2019a, 2019b). Version 2.0 adds a query language, expands the available algorithms, and provides tools for uncertainty analysis and recording how results were produced. It also includes a Model Context Protocol (MCP) server through which compatible assistants can access network analysis tools.

Statement of Need

Researchers in fields such as social science, biology, and physics often need to compare relationships across layers. A network of people, for example, might contain separate layers for work and friendship. Combining these layers into one graph can hide whether a connection belongs to one setting or both.

General graph libraries such as NetworkX (Hagberg et al., 2008) and igraph (Csardi & Nepusz, 2006) can store layer labels, but users must manage their interpretation. Dedicated multilayer packages offer data structures and analysis methods, while a researcher may still need to assemble separate tools for selecting layers, assessing uncertainty, and recording an analysis.

Py3plex represents each occurrence of a node in a layer separately and supports edges within and between layers. It offers a query language for selecting and analyzing parts of a network, along with tools for uncertainty analysis and provenance. Its methods include community detection, centrality measures, diffusion simulations, embeddings, and multilayer visualization.

For example, a researcher can use one query to select the work and friendship layers, calculate each node-layer pair's degree, and return the highest-degree nodes in each layer. Comparing the two lists can reveal people whose connections differ between settings. By default, degree counts all edges attached to a node-layer pair, including inter-layer edges; the researcher must account for those edges when interpreting the rankings.

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State of the Field

General graph libraries such as NetworkX (Hagberg et al., 2008) and igraph (Csardi & Nepusz, 2006) provide data structures and algorithms that can be used with layer labels. Dedicated multilayer packages include pymnet (Nurmi et al., 2024) and MultiNetX (Kouvaris, n.d.), both of which provide multilayer representations and operations. Pymnet also supports models with multiple aspects. MuxViz (De Domenico et al., 2015) provides multilayer analysis and visualization in R; Tulip (Auber et al., 2017) offers general graph visualization and analysis with Python support.

Py3plex 2.0 extends the existing Py3plex library, retaining its NetworkX-based representation and API. This allows its new query and uncertainty features to work with networks built using the earlier library. Researchers can choose among these tools according to their needs, such as pymnet's support for multiple aspects or muxViz's visualization tools.

Software Design

Py3plex represents a node in a layer as a pair `(node_id, layer_id)`. Its core network stores these pairs and their edges in a NetworkX MultiGraph or MultiDiGraph. This choice gives users access to NetworkX data structures and algorithms without requiring a dense tensor for the network. It also means that a node's role can be studied separately in each layer.

The library supports multilayer networks whose node sets differ across layers and multiplex networks in which nodes have counterparts across layers. Inter-layer edges in a general multilayer network are specified by the user. For multiplex networks loaded through the network loader, Py3plex can create coupling edges between replicas of the same node; these generated edges have a distinct type tag.

NetworkX interoperability has a trade-off. Some analyses operate directly on the multilayer representation, whereas others restrict the network to a layer or project it to a single graph before applying a delegated algorithm. For example, a degree calculated on a single layer counts connections within that layer, while a degree calculated on the full network can also count inter-layer connections. The execution path therefore matters when interpreting a result. For matrix-based analysis, Py3plex can also construct a supra-adjacency matrix, with within-layer connections on diagonal blocks and inter-layer connections on off-diagonal blocks. For performance, the library includes vectorized layer aggregation and a query planner that can reorder stages of a query.

Queries are represented internally as an abstract syntax tree, which can be inspected before execution. In supported workflows, the result can retain the query and its provenance alongside uncertainty estimates. The library also provides temporal network operations, network embeddings, and layouts that display layers and their inter-layer edges.

Research impact statement

Earlier versions of Py3plex have been used to visualize gene communities in a temporal study of grapevine infection (Škrlj et al., 2021) and to analyze and visualize networks in a comparison of nine languages (Škrlj & Pollak, 2019). Py3plex was also included in a comparative study of multilayer software that examined operator coverage and scalability (Panayiotou et al., 2024). These studies used earlier versions of the library.

Availability

Py3plex is released under the MIT licence. The repository also contains third-party Infomap code distributed under the GNU Affero General Public License version 3 or later. The source

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code, documentation and examples are available on GitHub, and the package is available on PyPI.

AI usage disclosure

The authors made the software design decisions and wrote the initial code and paper. OpenAI Codex assisted with coding and larger refactors. The authors used large language models to refine the paper and align its description with the codebase; Codex using GPT-6 assisted with the revisions in this submission. Earlier drafting and editing also used Google Gemini, OpenAI ChatGPT, and Anthropic Claude, and GitHub Copilot and Claude Code assisted with code and tests. The authors are responsible for reviewing and validating all AI-assisted material. # Acknowledgements

This work was partly funded by the Slovenian Research Agency under the projects: P2-0103, PR-12394 and GC-0002 (Large Language Models for Digital Humanities). This work was also partially supported by the European Union's Horizon Europe research and innovation program under grant agreement No. 101214398 (ELLIOT). Views and opinions expressed are, however, those of the author(s) only and do not necessarily reflect those of the European Union or the European Commission. Neither the European Union nor the European Commission can be held responsible for them. Furthermore, the authors thank the py3plex user and contributor community for feedback that has helped improve the library.

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